

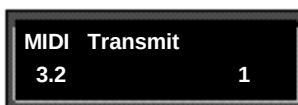
Controller quirks

Some synthesizers and controllers cannot send the full range of MIDI program change messages (1-128). Others may appear to be able to send only 32, but actually have a bank mode that *does* let you send all 128 program change messages. Also, be aware that some MIDI devices use a program numbering system that uses 0-127 instead of 1-128. If in doubt, see the manual for your controller.

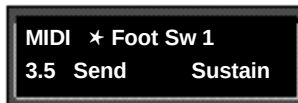
The ADJUST Knob, Foot Pedal, Foot Sw 1 and Foot Sw 2 as MIDI Controllers

You can choose to have the PCM 81 send MIDI Controller messages whenever you turn ADJUST (the soft knob in Program Banks or Register Banks modes), or to activate analog controllers connected to the rear panel Footswitch or Foot Controller jacks. This makes it possible to record real-time control of PCM 81 effects with a MIDI sequencer — A simple but quite powerful way to automate effects.

To send MIDI data from these controllers, first set Control mode 3.2 (**Transmit**) to the desired MIDI Channel. (The default is Channel **1**.)



Once a transmit channel has been set, go to Control mode 3.5 to assign MIDI Controllers. Press **Load/★** to display the available controllers: **Foot Pedal**, **Foot Sw1**, **Foot Sw2** or **ADJUST**. Turn ADJUST to assign the MIDI Controller data to be sent when the displayed controller is activated. (The default assignment is **None**.).



When a PCM 81 controller is assigned to a MIDI Controller, the PCM 81 will respond to incoming controller messages as though its own controller were moved. In other words, if **FootSw1** is assigned to **Sustain**, the PCM 81 will respond to incoming Sustain messages as though FootSw1 had been activated.

Each PCM 81 preset has a unique soft knob patch that allows you to control the effect directly from Program or Register Banks mode with the ADJUST knob. You can also control the soft knob patch remotely from MIDI, or from the Foot Pedal.

Controlling the Soft Knob with MIDI

To control the soft knob with MIDI, set Control mode 3.1 (**Receive**) to the desired MIDI Channel. Set Control mode 3.5 (**ADJUST**) to the desired MIDI Controller such as **Mod Wheel**.

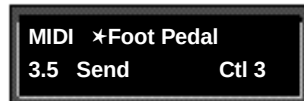
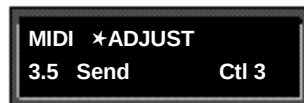


Now, the Mod Wheel on the MIDI instrument will control the soft knob patch of the running effect.

If you have a foot pedal connected to the PCM rear panel Foot Controller jack, you can use it to control the soft knob patch. (Note that no MIDI connections are required to do this.)

Controlling the Soft Knob with a Foot Pedal

Set both Control mode 3.5 **ADJUST** and Control mode 3.5 **Foot Pedal** to the same MIDI Controller.



Now, the foot pedal will control the soft knob patch of the running effect.

Program Change Messages

Reception of MIDI Program Change and Bank Select messages can be selectively enabled/disabled from Control Mode parameter 3.0, MIDI Pgm Change. The manner in which the PCM 81 interprets these messages is determined by the value of this parameter as follows:

Pgm Change: Off

All Program Change and Bank select messages are ignored. Pgm+ and Pgm– will load the next higher or lower program in the current bank.

Pgm Change : On

Program Change messages 0-49 correspond to PCM 81 Effects 0.0 -4.9 in the current bank. Program Change messages 50—127 are ignored. Pgm+ and Pgm– will load the next higher or lower program in the current bank.

The current bank can be changed with MIDI Bank Select Messages as follows:

- 0-5: Program Banks 0-5
- 6: Internal Register Bank
- 7-11: reserved
- 12-58: Memory Card Banks

The number of banks available on a given card will vary with card size as follows:

Card Size	# Banks
64	2
256	11
512	23
1 Meg	47

Pgm Change: Map

Program Change 0-127 can be mapped to any PCM 81 Effect in any internal or card bank. Two 128 element maps are stored internally, additional maps may be stored on RAM cards.

Map 0	Map 1
MIDI 0 = P0 0.0	MIDI 0 = P2 2.8
MIDI 127 = P2 2.7	MIDI 127 = R 4.9

Pgm+ and Pgm– will load the next higher or lower program in the map.

Pgm Change: Chain

Any Program Change number can be selected to load any one of ten customized effect “chains.” Once a chain is loaded, effects in the chain are accessed by the controller patched to Pgm + and Pgm – (program increment and program decrement).

Automation

SysEx Automation

The PCM 81 will transmit SysEx automation messages when Control Mode parameter 3.4, MIDI Automation is set to On. All changes made by front panel operations are transmitted as PCM 80 SysEx messages. This is intended primarily for use by editor programs and in configurations where it is desirable for one or more PCM 81s to be slaved to a single PCM 81 acting as a master. The current mode (Program Banks, Register Banks, Edit, Control or Tempo) of the slave does not follow the master, but the actual parameter values do. (SysEx automation can also be stored on a sequencer and replayed in real-time. As a general rule, automating more than two or three SysEx program changes at once is not recommended.) Because messages are transmitted as PCM 80 messages, either PCM 80s or PCM 81s can be slaved to the master.

The PCM 81 can receive SysEx messages when Control mode 3.7 MIDI SysEx is set to **Receive On**. Note that when using SysEx automation, the device of the receiving PCM 81 must match the Target ID of the transmitting PCM 81.

The Target ID setting can be adjusted from Control 3.4 MIDI Automation. It is selected by pressing **Load/★** after setting Automation to **On**. The default setting for the Target ID is **All**.

The Device ID setting can be adjusted from Control mode 3.7 MIDI SysEx. It is selected by pressing **Load/★** after setting SysEx to **Receive On**. The default setting for the Device ID is **0**.

Controller Automation

For applications where it is desirable to "automate" changes made to PCM 81 effects with its own controls (ADJUST knob, Foot Pedal, Footswitch 1 or Footswitch 2), we recommend assigning the controllers to MIDI destinations and recording the changes with a MIDI sequencer (see Control Mode parameter 3.5, MIDI Destinations).

Reset All Controllers

The PCM 81 recognizes the "Reset All Controllers" message. When received, all patched parameters are reset to their stored values. Patched parameters may also be reset from the PCM 81 front panel —Control Mode parameter 3.0 (the message will be transmitted from the PCM 81 as well).

MIDI Clock and Clock Commands

The PCM 81 recognizes MIDI clock messages when Tempo Mode parameter 0.2, Tempo Source is set to MIDI. Any Delay or LFO parameter set to display tempo values will be synchronized to the tempo of the incoming MIDI clock.

MIDI Clock and Clock Commands are also available as Dynamic MIDI patch sources. The value of MIDI Clock when used as a patch source is a linear scaling of 0 to 127 (0 = 40 BPM and 127 = 400 BPM). The value of Clock Commands when used as a Dynamic MIDI patch source is 1 for START and CONTINUE and 0 for STOP.

PCM 80 Compatibility

The PCM 81 can receive MIDI data from either the PCM 80 or another PCM 81. It can also transmit certain messages in PCM 80 format. (See *Bulk Data Dumps* and *SysEx Automation*.) This allows most data to be exchanged between the two products, with any format translations handled automatically by the PCM 81.

Dynamic MIDI

The following MIDI messages are available as Dynamic MIDI patch sources:

- MIDI Controllers 1-119
- Pitch Bend
- After Touch (Polyphonic and Channel combined)
- Velocity (Note On)
- Last Note
- Low Note
- High Note
- Tempo (40–400BPM is converted to controller range 0-127)
- Clock Commands

These MIDI messages are also available as threshold sources for several Modulation parameters: AR Env, Latch, Sw 1 and Sw 2. They may also be used as a tap source for controlling Tempo.

MIDI Implementation Details, including System Exclusive documentation, are available to assist experienced programmers in developing software for use with the PCM 81. These can be obtained directly from Lexicon.
Request: PCM 81 MIDI Implementation Details.

Bulk Data Dumps

Control mode 3.8 (**MIDI Dump**) allows selection of the following types of bulk data to be dumped directly from the PCM 81 to another PCM 81, or to editor/librarian software.

Displayed Name	Description
CurrentPgm*	Currently running effect
Bank R*	Internal Register Bank
Bank C0-CK*	Card Banks (card must be inserted)
Map 0, 1*	Internal Program Change Maps
Map 2-33*	Card Program Change Maps (card must be inserted)
Chain 0-9*	Internal Program Chains
Chain 10-19*	Card Program Chains (card must be inserted)
Int Chains*	All Internal Program Chains
Ext Chains*	All Card Program Chains (card must be inserted)
Setup C	Current Setup
Setup 0-4	Internal Setups
Setup 5-9	Card Setups

* Transmitted in PCM 80 format

Use ADJUST to select the bulk data type. Press **Store** to transmit the data .

MIDI Implementation Chart

Lexicon PCM 81
Digital Effects System

Function		Transmitted	Recognized	Remarks
Basic Channel	Default Changed	1 1-16	1 1-16	
Mode	Default Messages Altered	X X	Mode 1, 3 X X	
Note Number		X	0-127	Last Note, Low Note, High Note used as controllers
Velocity	Note ON Note OFF	X	O 9n v = 1-127	used as controller
After Touch	Keys Channel	X X	X O	
Pitch Bend		X	O	
Control Change	1-119	OX	OX	ADJUST, Footpedal, Footswitch 1, and Footswitch 2 can be assigned controllers 1-119 for MIDI transmit
Program Change	True #	X	0-127	See Implementation Details
System Exclusive	Lexicon Real-time non Real-time	OX X X	OX X OX	mfgr ID=6; product ID=7 device ID
System Common	:Song Pos :Song Sel :Tune	X X X	X X X	
System Real Time	:Clock :Commands	OX X	OX OX*	START, STOP and CONTINUE are patchable as a switch: START/CONTINUE=127; STOP=0
Aux Messages	:Local ON/OFF :All Notes OFF :Active Sense :Reset All Controllers	X X X OX	X O X OX	
Notes: PCM 81 transmits and receives in both PCM 80 (product ID 0x07) and PCM 81 (product ID 0x10) formats.				

Mode 1: OMNI ON, POLY
Mode 3: OMNI OFF, POLY

Mode 2: OMNI ON, MONO
Mode 4: OMNI OFF, MONO

O : Yes **OX: Selectable**
X : No

Troubleshooting

This chapter is intended primarily to help you recognize some common error states which can be corrected from the PCM 81 front panel, or by simple means such as cable replacement. Any error states which are not covered here should be referred to your local dealer for service by a qualified technician.

In a low-voltage, or "brown-out" condition, the PCM 81 will freeze in its current state. None of the controls will have any effect. When power returns to a normal level, the unit will reset itself as though it had just been powered on. If the unit does not reset itself, turn the power OFF, then ON to resume normal operation.

Low Voltage

Temperature extremes may cause the PCM 81 to exhibit unpredictable behavior. If the unit has been subjected to temperatures below 32°F (0°C) or above 95°F (35°C), it should be turned off and allowed to return to normal temperature before use. The unit may be damaged by exposure to temperatures below -22°F (-30°C) or above 167°F (75°C), or by exposure to humidity in excess of 95%. If a unit exposed to such conditions fails to operate after it returns to a normal operating temperature, contact your local service representative.

Overheating

The PCM 81 doesn't respond to MIDI Program Changes.

Check **Receive** control at Control mode 3.1 and make sure it is set to **On**. Check the MIDI Channel selected as well as the MIDI Channel of the transmitting device. Make sure that **Pgm Change** at Control mode 3.3 is set to **On**. Also check MIDI In/Out connections between the units.

Common MIDI Problems

MIDI Program Change numbers are off by 1.

The PCM 81 transmits and recognizes ProgramChange messages 0-127. MIDI devices which transmit 1-128 rather than 0-127, will be off by 1. Simply adjust by 1 when working with such devices.

The PCM 81 doesn't respond to SysEx commands.

Check the **SysEx** setting (and the Device ID selection) at Control mode 3.7.

The PCM 81 does not transmit SysEx Automation commands.

Check the **Automation** setting (and the target device ID selection) at Control mod 3.4.

- Operational Problems** **The PCM 81 will not lock onto an incoming digital signal.**
Check the cables that you are using. DO NOT USE ANALOG AUDIO CABLE TO CONNECT DIGITAL AUDIO.
- Also check to make sure that your input signal complies with S/PDIF format standards. The PCM 81 will recognize AES professional format signals from an appropriate connector, but will not necessarily read and transmit encoded information accurately.
- No Input**
Analog: Check the analog input connection and make sure that the **Analog Lvl** control at Control mode 0.1 is set to 100%.
- Digital AES (XLR):** Check connection to XLR and make sure that the **Digital Lvl** control at Control mode 0.2 is set to 100%. Make sure that **WordClock** (Control mode 0.0) is set to **Ext: XLR 48 or 44.1**.
- Digital S/PDIF (Coax):** Check connection to Coax and make sure that the **Digital Lvl** control at Control mode 0.2 is set to 100%. Make sure that **WordClock** (Control mode 0.0) is set to **Ext: XLR 48 or 44.1**.
- No Digital Audio Output**
Check the **Analog Lvl** and **Digital Lvl** controls at Control mode 0.1 and 0.2.
- No Effects Output**
Check the setting of Mix Mode at Control mode 1.1. Also verify that any controllers patched to **FX Lvl**, **Input Level**, or **Mix** are not turned off.
- Power On Behavior** The PCM 81 performs a series of self tests each time it is powered on, then displays the PCM 81 copyright notice. This should be followed by the display and loading of the last loaded effect. If this sequence does not occur, contact Lexicon Customer Service.

You can restore the PCM 81 to its default state *without* erasing registers by restoring the factory default setup:

Restoring Factory Default Settings

1. Press **Control**.
2. Use the **Up** and **Down** buttons to locate **Row 4 Setup**.
3. Turn SELECT to **4.1 Load**.
4. Turn ADJUST counterclockwise to select "Factory Settings".
5. Press **Load/★**. The PCM 81 will display the message "Setup restored".

The following table shows the parameters which comprise a setup, along with the factory default setting of each parameter.

Control Mode		System Parameter	Default Setting
Matrix Location			
Audio	0.0	Word Clock	Internal 48kHz
	0.1	Analog Lvl	100%
	0.2	Digital Lvl	0%
	0.3	Word Size	20 bits
	0.4	SCMS	Multi Copy
	0.5	Emphasis Bit	Pass Thru
	0.6	Output Level	+4dBu
System	1.0	Edit Mode	Go
	1.1	Mix Mode	Pgm
		Global Mix Value	100% Wet
	1.2	Tempo Mode	Pgm
		Global Tempo Value	120 BPM
	1.3	Bypass Mode	InputMute
		Bypass Src	Off
	1.4	Pgm Bypass	AllMute
	1.5	Mem Protect	On
	1.6	Auto Load	Off
	1.7	Patch Update	Delayed
MIDI	3.1	Receive	OMNI
	3.2	Transmit	1
	3.3	Pgm Change	On
		Pgm+	Off
		Pgm-	Off
		Map select	0
		Chain	MIDI
	3.4	Automation	Off
	3.5	Footpedal	None
		Sw 1	None
		Sw 2	None
		ADJUST	None
	3.6	Int Clock	Off
	3.7	SysEx	On
		Device ID	0
	3.9	Dump Speed	Slow
Tempo Mode		System Parameter	Default Setting
Matrix Location			
Tempo	0.2	Source	Internal
Tap	1.3	Display	On

Reinitialization**Reinitializing will erase all registers and setups**

The following procedure will return the PCM 81 to the state it was in when shipped from the factory. This includes erasing all registers and setups, as well as restoring all of the default settings:

1. Press **Control**.
2. Use the **Up** and **Down** buttons to locate **Row 1 System**.
3. Turn SELECT to **1.8 Initialize**.
4. Press **Store**. The PCM 81 will display the message "Are you sure? (Press STORE).

If you don't want to reinitialize your unit, press any button *except* **Store** to return to matrix position 1.8.

If you press **Store** in response to this message, the display will flash "Restoring original factory settings" and your unit will be reinitialized.

PCM 81 Specifications

Audio Input	Connectors:	Combined 3 pole XLR and 1/4 inch T/R/S phone jacks (2)
	Impedance:	0 dB/BAL switch position: 100k Ω , balanced -20 dB/UNBAL switch position: 50k Ω , unbalanced
	Levels:	0 dB/BAL switch position: -2 dBu min for full scale, +20 dBu max -20 dB/UNBAL switch position: -22 dBu min for full scale, 0 dBu max
	CMRR:	0 dB/BAL switch position: 50 dB minimum, 10 Hz to 20 kHz
Audio Output	Connectors:	1/4 inch T/R/S phone jacks (2); balanced XLRs, pin 2 "high" (2)
	Impedance:	100 Ω , balanced
	Levels:	+18 dBm, full scale (+4 dBu setting) balanced, unbalanced +4 dBm, full scale (-10 dBu setting)
	Protection:	Relays provided for output muting during power on/off
A/D Performance	Frequency Response:	10 Hz to 20 kHz, ± 0.5 dB
	Crosstalk:	<-65 dB, 10 Hz to 20 kHz
	S/N Ratio:	>102 dB, 20 kHz bandwidth
	THD:	<0.003% max, 10 Hz to 20 kHz
	Dynamic Range:	>102 dB, 20 kHz bandwidth
	Delay:	24 samples (0.54 msec for 44.1 kHz, 0.50 msec for 48 kHz)
D/A Performance	Frequency Response:	10 Hz to 20 kHz, ± 0.5 dB
	Crosstalk:	<-80 dB, 10 Hz to 20 kHz
	S/N Ratio:	>98 dB, 20 kHz bandwidth
	THD:	<0.005%, 10 Hz to 20 kHz
	Dynamic Range:	>98 dB, 20 kHz bandwidth
	Delay:	50 samples (1.13 msec for 44.1 kHz, 1.04 msec for 48 kHz)
A/A Performance	Frequency Response:	10 Hz to 20 kHz, ± 0.5 dB
	Crosstalk:	<-55 dB, 10 Hz to 20 kHz
	S/N Ratio:	>96 dB, 20 kHz bandwidth
	THD:	<0.005%, 10 Hz to 20 kHz
	Dynamic Range:	>96 dB, 20 kHz bandwidth

Digital Audio Interface	Connectors:	Coaxial, RCA type (2); balanced, XLR (2)
	Format:	S/PDIF (IEC-958) and AES/EBU (AES3-1992) professional interface
	Sample Rates:	44.1 kHz, 48 kHz
Internal Audio Data Paths	Conversion:	20 bits
	DSP:	20 to 24 bits
External Memory Card	Connector:	Accepts PCMCIA Type I cards, 68 pins
	Standards:	Conforms to PCMCIA 2.0 / JEIDA 4.0
	Card Format:	Supports up to 1MB SRAM (attribute memory not required)
Control Interface	MIDI:	5-pin DIN connectors provided for MIDI IN, THRU, & OUT
	Footswitch:	1/4 inch T/R/S phone jack provided for 2 independent momentary footswitches System detects normal-open, or normal-closed on power up
	Foot controller:	1/4 inch T/R/S phone jack provided for footpedal (100 Ω minimum, 10k Ω maximum impedance)
General	Dimensions:	19.0"W x 1.75"H x 12.0"D (483 x 45 x 305 mm) 19 inch rack mount standard, 1U high
	Weight:	Net: 6.4 lbs (2.9 kg) Shipping: 9.5 lbs (4.3 kg)
	Power	
	Requirements:	100-240 VAC, 50-60 Hz, 35 W, 3-pin IEC power connector
	RFI/ESD:	Conforms to FCC Class B, EN55022 Class B (CE), IEC 801-2, IEC 801-3
	Environment:	Operating temperature: 32° to 104°F (0° to 40°C) Storage temperature: -22° to 167°F (-30° to 70°C) Humidity: maximum 95% without condensation

Unless otherwise noted, all audio specifications assume rear-panel switch set to BAL, input level control is set for unity gain (0dB), and analog I/O connections wired for balanced configuration.

Specifications subject to change without notice.

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